

GUJARAT TECHNOLOGICAL UNIVERSITY
BE-MINOR- SEMESTER-IV EXAMINATION – SUMMER 2022

Subject Code:114AG01**Date:13-07-2022****Subject Name:Introduction to AI and Machine Learning****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

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|------------|------------|---|-----------|
| Q.1 | (a) | Explain the significance of Sigmoid function in logistic regression. | 03 |
| | (b) | List any four differences between supervised and unsupervised learning. | 04 |
| | (c) | Discuss any four applications of Machine learning. | 07 |
| Q.2 | (a) | Explain the following terms with an example: | 03 |
| | | 1. Random experiment | |
| | | 2. Discrete and continuous distributions | |
| | (b) | Discuss regression and its significance. List any four regression techniques. | 04 |
| | (c) | List the steps of machine learning process and show the flow between steps. Explain steps in detail. | 07 |
| | | OR | |
| | (c) | Explain mutli-classification (one vs all) in detail with example. | 07 |
| Q.3 | (a) | What is supervised learning? Which are the different types of supervised learning algorithms? | 03 |
| | (b) | When and why overfitting occurs? How it can be overcome? | 04 |
| | (c) | Explain logistic regression in detail. Which are the different types of logistic regression? | 07 |
| | | OR | |
| Q.3 | (a) | What is unsupervised learning? Which are the different types of unsupervised learning algorithms? | 03 |
| | (b) | Discuss how matrices are useful in machine learning? Discuss common matrix operations. | 04 |
| | (c) | What is procedural knowledge? What is declarative knowledge? List any three differences between them. Also give an example of procedural knowledge and declarative knowledge. | 07 |
| Q.4 | (a) | List any three differences between linear regression and logistic regression. | 03 |
| | (b) | What is classification problem? Explain the different types of classification. | 04 |
| | (c) | Discuss simple linear regression in detail. | 07 |
| | | OR | |
| Q.4 | (a) | List any three differences between clustering and classification. | 03 |
| | (b) | State any four differences between regression and classification. | 04 |
| | (c) | Define Artificial Intelligence. Discuss its significance and applications in today's era. Briefly discuss the different AI techniques. | 07 |
| Q.5 | (a) | What is cost function in machine learning? Why is it required? | 03 |

(b) What is clustering? Explain with example. List any two applications of clustering. **04**

(c) Discuss any one clustering technique in detail with example. **07**

OR

Q.5 (a) What is gradient decent in machine learning? Explain with diagram. **03**

(b) Explain with example that how predicate logic is useful to represent knowledge? **04**

(c) 1. Write a prolog program to drop every nth element from the list. **07**

2. Write a prolog program to determine whether given number is prime or not?
